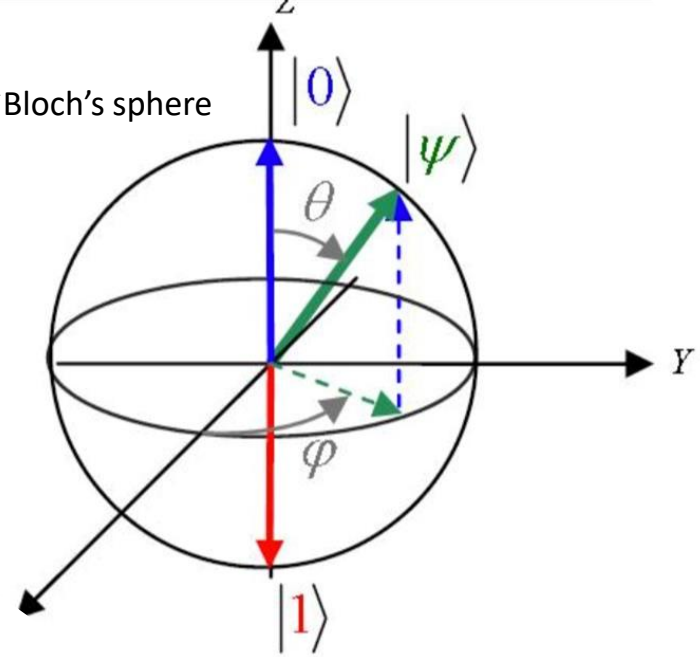
Physics of Computation Conference jointly organized by MIT and IBM, and held at the MIT Endicott House in 1981

« Quantum Computing, more than a dream »

Eric MELY,
Flex leader Quantum Computing
HUB Emerging Technologies

`Bloch's sphere



Quantum Computing?

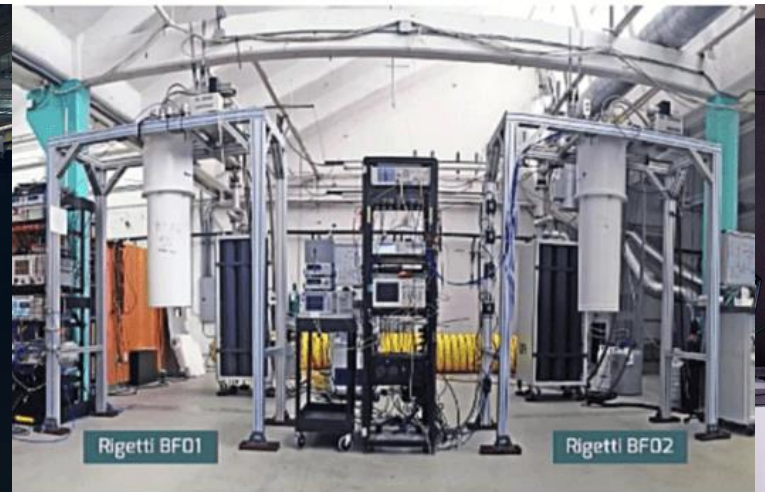
- **"Nature is quantum, goddamn it! So if we want to simulate it, we need a quantum computer."**
- **R.Feynman, 1981, Endicott House, MIT**

Where it all began!

1- Quantum computers

What is a Quantum Computer ?

A quantum computer is a machine that performs **calculations based on** the **laws of quantum mechanics**, which is the behavior of particles at the **sub-atomic level.**



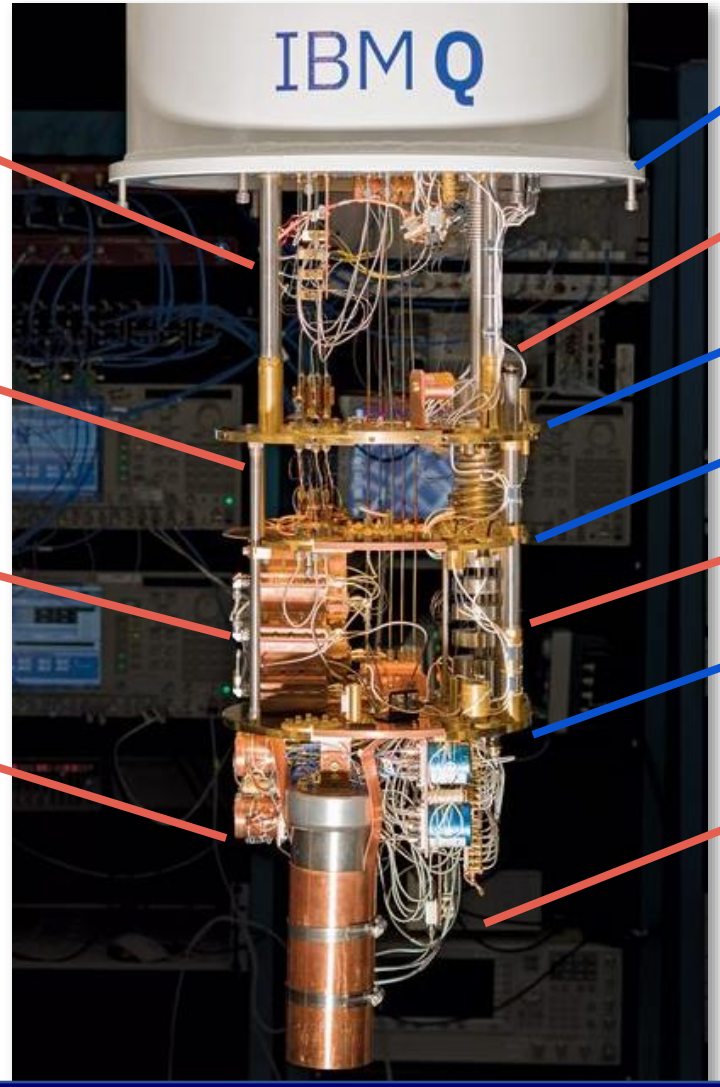
IBM Q, Inside a Quantum Computer

Qubit signal amplifier

Superconducting coaxial lines

Cryogenic isolators enable qubit signals to go forward while preventing noise from compromising qubit quality

Quantum amplifiers, inside of a magnetic shield capture and amplify processor readout signals while minimizing noise



The march to aboslute zero
4 Kelvins

Input microwave lines

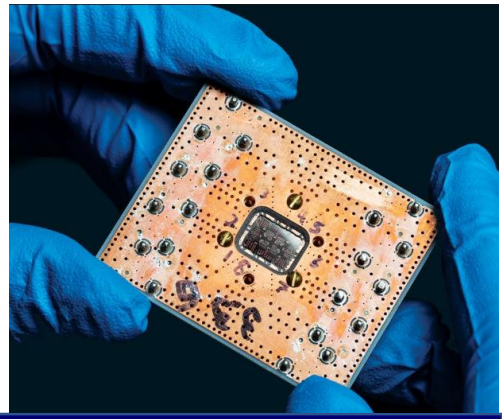
800 milli Kelvins

100 milli Kelvins

Mixing chamber

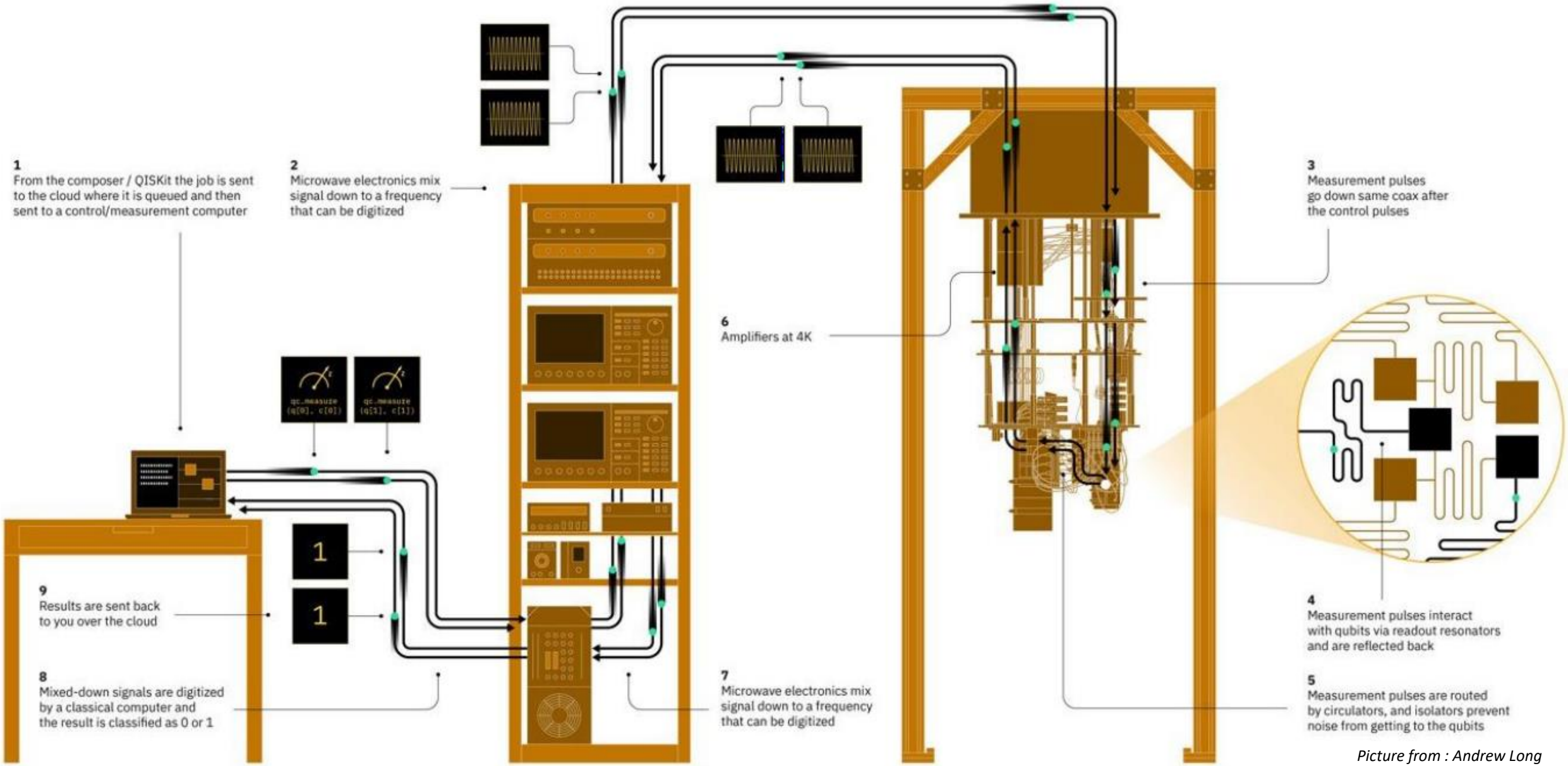
15 milli Kelvins

Quantum CPU, inside a shiled that protects it from electromagnetic radiation



Harnessing the power of a quantum processor requires maintaining constant temperature near absolute zero, and protect it from external electromagnetic radiation and physical vibration.

Around a Quantum Computer



Picture from : Andrew Long

What makes the difference ?

Superposition

+

Entanglement

=

***Power* of a Quantum computer**

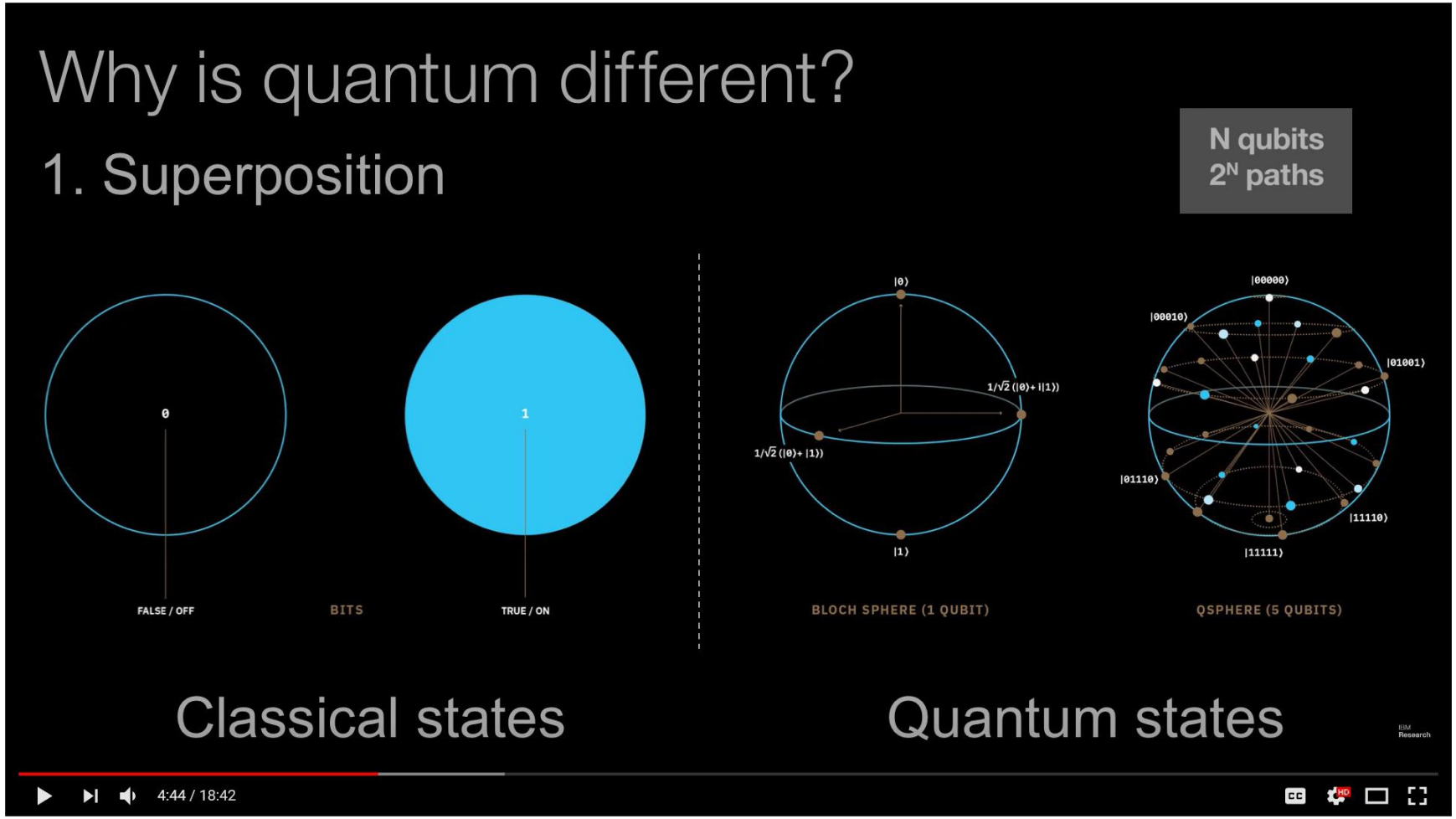
2^N states with N qubits

300 qubits can encode more states than atoms in the univers!

Contrains

No-copy theory

Measurement



A Beginner's Guide to Quantum Computing

172,177 views

2.7K 99 SHARE

IBM Research
Published on May 31, 2017

https://www.youtube.com/watch?v=R6iWFmLC2-E

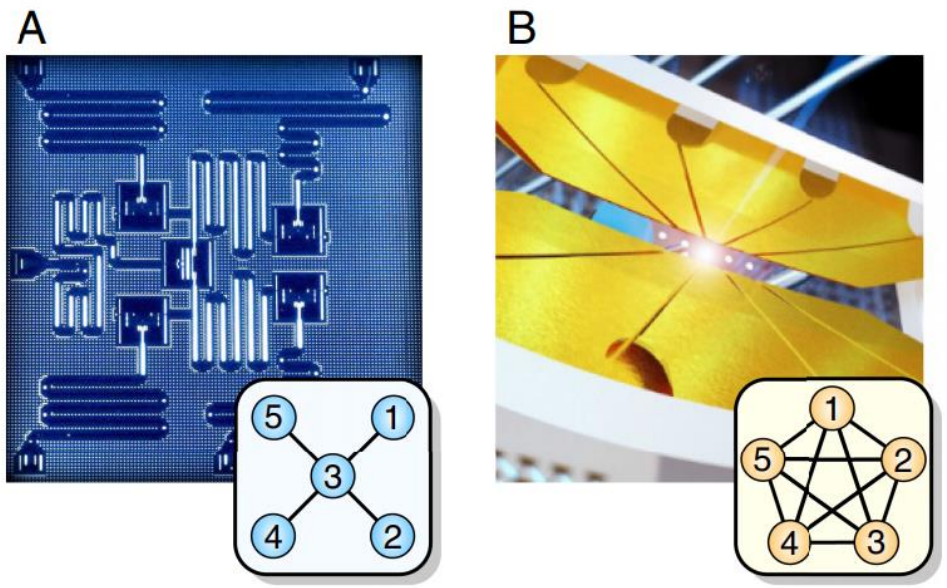
SUBSCRIBE 27K

Multiple qubit technologies

Most Quantum computer are based on

A - Superconducting Josephson Junction

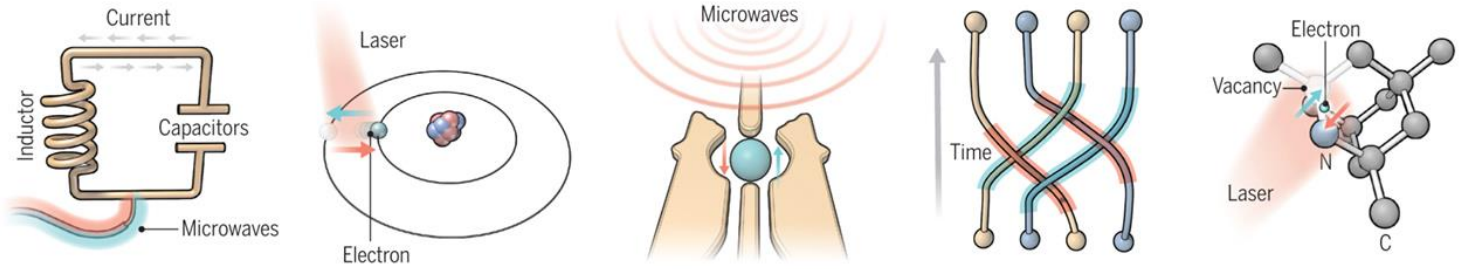
B - Trapped ion



Graphic representations of two 5 qubits systems. (A) the superconducting qubits connected by microwave resonators (credit: IBM Research). (B) The linear chain of trapped ions connected by laser-mediated interactions (Honeywell). (A and B, Insets) Qubit connectivity graphs: (A) star shaped and (B) fully connected.

A bit of the action

In the race to build a quantum computer, companies are pursuing many types of quantum bits, or qubits, each with its own strengths and weaknesses.



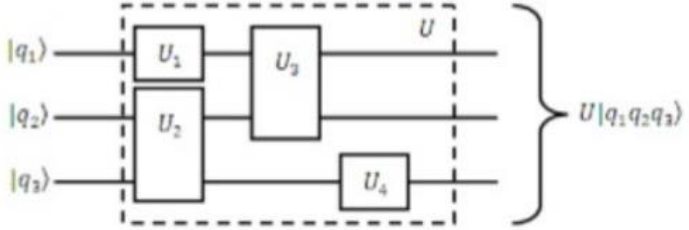
	Superconducting loops	Trapped ions	Silicon quantum dots	Topological qubits	Diamond vacancies
	A resistance-free current oscillates back and forth around a circuit loop. An injected microwave signal excites the current into superposition states.	Electrically charged atoms, or ions, have quantum energies that depend on the location of electrons. Tuned lasers cool and trap the ions, and put them in superposition states.	These "artificial atoms" are made by adding an electron to a small piece of pure silicon. Microwaves control the electron's quantum state.	Quasiparticles can be seen in the behavior of electrons channeled through semiconductor structures. Their braided paths can encode quantum information.	A nitrogen atom and a vacancy add an electron to a diamond lattice. Its quantum spin state, along with those of nearby carbon nuclei, can be controlled with light.
Longevity (seconds)	0.00005	>1000	0.03	N/A	10
Logic success rate	99.4%	99.9%	~99%	N/A	99.2%
Number entangled	9	14	2	N/A	6
Company support	Google, IBM, Quantum Circuits	ionQ	Intel	Microsoft, Bell Labs	Quantum Diamond Technologies
Pros	Fast working. Build on existing semiconductor industry.	Very stable. Highest achieved gate fidelities.	Stable. Build on existing semiconductor industry.	Greatly reduce errors.	Can operate at room temperature.
Cons	Collapse easily and must be kept cold.	Slow operation. Many lasers are needed.	Only a few entangled. Must be kept cold.	Existence not yet confirmed.	Difficult to entangle.

Note: Longevity is the record coherence time for a single qubit superposition state, logic success rate is the highest reported gate fidelity for logic operations on two qubits, and number entangled is the maximum number of qubits entangled and capable of performing two-qubit operations.

Picture from : Science, Dec 2016, Vol 354, Issue 6316

Two Quantum Computer models

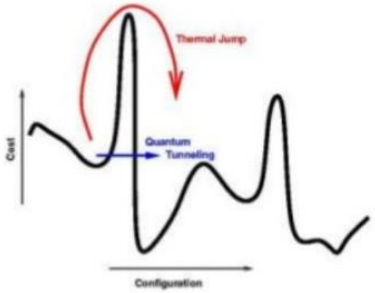
Circuit Model – Universal Computer



- Digital
- Logical Gates
- Predictable behaviour at scale
- “Mainstream” approach
- **IBM, Google, Righetti, Intel, Microsoft, ATOS***

NISQ (Noisy Intermediate-Scale Quantum) Computer (less then 1000 qubits)

Adiabatic Model – Specialized Computer

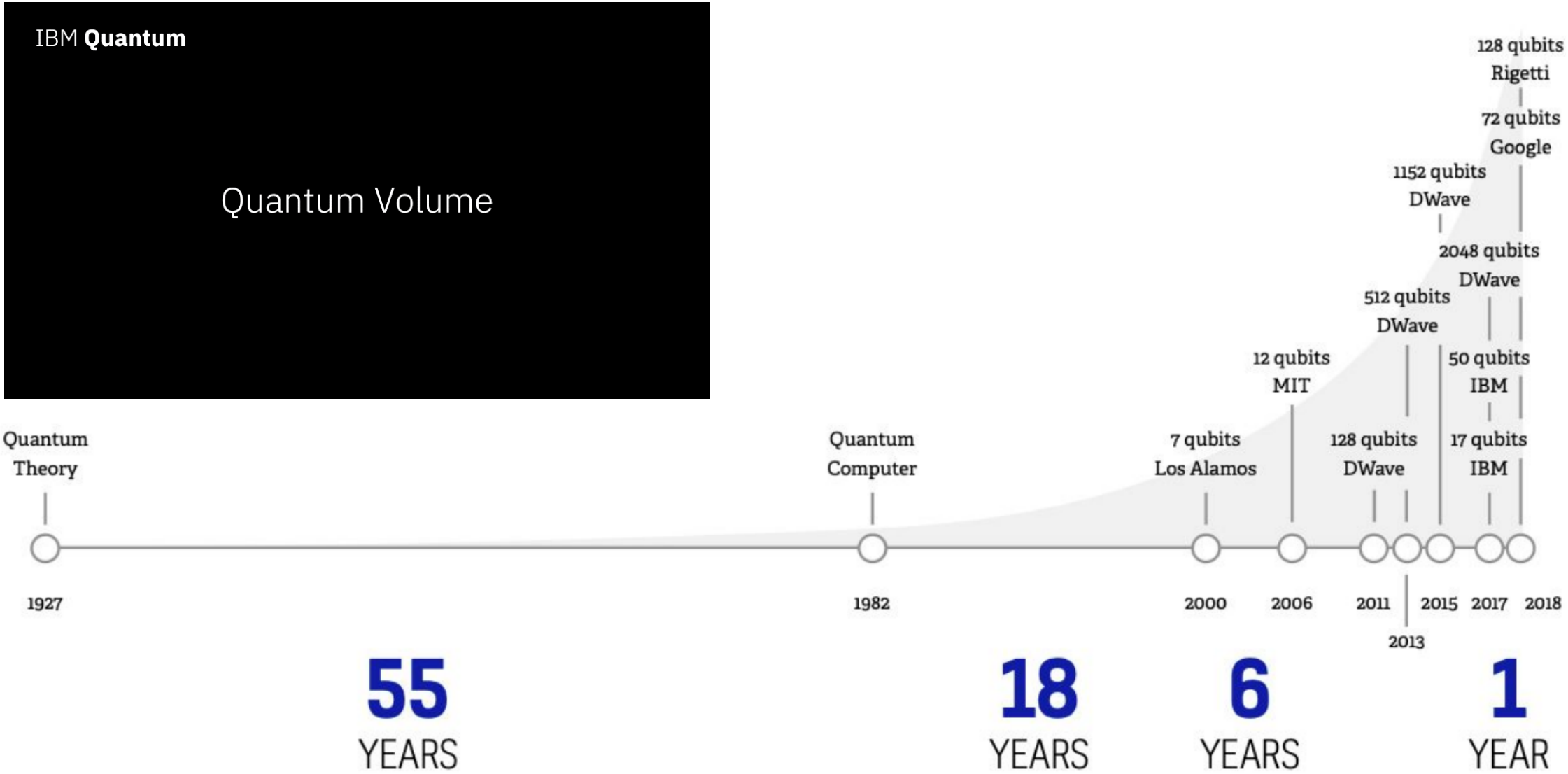


- Analogic
- Math problem solved physically
- Solutions are low energy states
- Hard to predict behavior at scale
- No error correction
- **D-Wave, Fujitsu*, Toshiba***

**emulators*

Quantum hardware development is accelerating

1000 qubits for 2023 (IBM)



Source: StrangeWorks

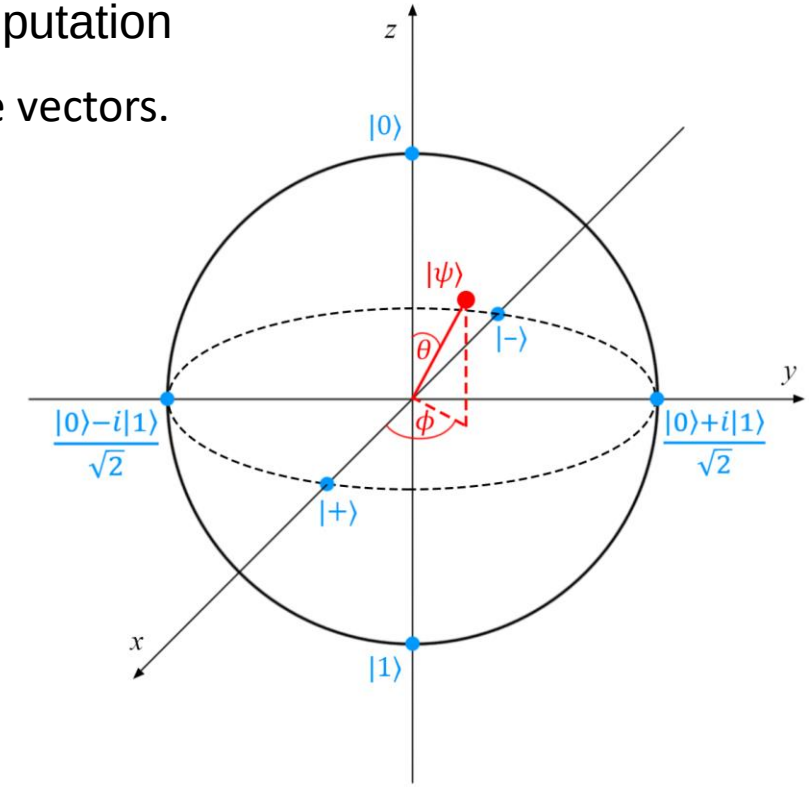
Quantum bits *(qubit)*

- a "qubit" stores the **probability of information**. It represents both "1" and "0" at the same time.
- **All the point** on the Bloch sphere are part of the quantum computation
- A quantum computer is a machine that **simply rotates** quantum state vectors.
- $\mathbf{2^n}$ **dimensional** vector needs only n quantum bits.
- Measurement **gives $\boldsymbol{n}$ bits** of information.
- The representation is based on **a matrix** (dirac notation):

$$|\psi\rangle = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}$$

$$|\psi\rangle = c_0|0\rangle + c_1|1\rangle$$

$$|\psi\rangle = cos(\theta)\,|0\rangle + e i^{\phi} sin(\Theta)\,|1\rangle$$



Bloch Sphere

Quantum Gates, all about matrix!

Single qubit states

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$$

$$|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Bell States/EPR pairs - simplest examples of quantum entanglement between two qubits:

$$|\psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

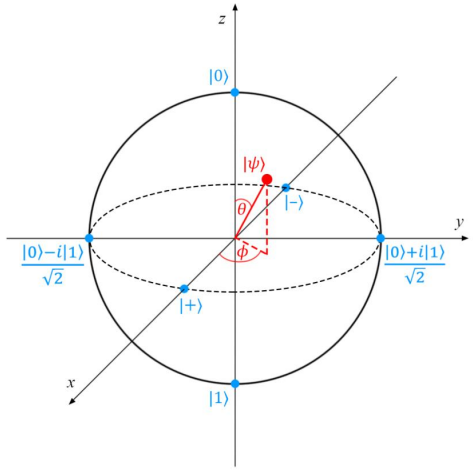
$$|\phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

The GHZ (Greenberger-Horne-Zeilinger) states, in general form (for n qubits) and in its simplest (three-qubit) form – 'maximal entanglement'

$$|GHZ\rangle_{general} = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$$

$$|GHZ\rangle_{simplest} = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

Name(s)	Matrix	Circuit Symbol(s)	Q# Representation	Key Operations
Pauli *X*, *X*, NOT, bit flip, σ_x	$\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$	X	X(qubit : Qubit)	$X\|0\rangle = \|1\rangle$ $X\|1\rangle = \|0\rangle$ $X\|+\rangle = \|+\rangle$ $X\|-\rangle = -\|-\rangle$
Pauli *Y*, *Y*, σ_y	$\begin{bmatrix}0 & -i\\i & 0\end{bmatrix}$	Y	Y(qubit : Qubit)	$Y\|0\rangle = i\|1\rangle$ $Y\|1\rangle = -i\|0\rangle$ $Y\|+\rangle = -i\|-\rangle$ $Y\|-\rangle = i\|+\rangle$
Pauli *Z*, *Z*, phase flip, σ_z	$\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$	Z	Z(qubit : Qubit)	$Z\|0\rangle = \|0\rangle$ $Z\|1\rangle = \|1\rangle$ $Z\|+\rangle = \|-\rangle$ $Z\|-\rangle = \|+\rangle$



Measure operator

The Pauli matrices are their own inverse:

$$X^2 = Y^2 = Z^2 = \mathbb{I}$$

Hadamard, *H*	$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$	H	H(qubit : Qubit)	$H\|0\rangle = \|+\rangle$ $H\|1\rangle = \|-\rangle$ $H\|+\rangle = \|0\rangle$ $H\|-\rangle = \|1\rangle$
Phase Shift, R_ϕ	$\begin{bmatrix}1 & 0\\0 & e^{i\phi}\end{bmatrix}$	R_ϕ	R1(theta : Double, qubit : Qubit) More generally: R(pauli : Pauli, theta : Double, qubit : Qubit)	$R_\phi\|0\rangle = \|0\rangle$ $R_\phi\|1\rangle = e^{i\phi}\|1\rangle$ $R_\phi\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + e^{i\phi}\|1\rangle)$ $R_\phi\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - e^{i\phi}\|1\rangle)$
Phase, $\frac{\pi}{4}$, *S*	$\begin{bmatrix}1 & 0\\0 & i\end{bmatrix}$	S	S(qubit : Qubit)	$S\|0\rangle = \|0\rangle$ $S\|1\rangle = i\|1\rangle$ $S\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + i\|1\rangle)$ $S\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - i\|1\rangle)$
$\frac{\pi}{8}$, *T*	$\begin{bmatrix}1 & 0\\0 & e^{i\frac{\pi}{4}}\end{bmatrix}$	T	T(qubit : Qubit)	$T\|0\rangle = \|0\rangle$ $T\|1\rangle = e^{\frac{i\pi}{4}}\|1\rangle$ $T\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + e^{\frac{i\pi}{4}}\|1\rangle)$ $T\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - e^{\frac{i\pi}{4}}\|1\rangle)$

SWAP	$\begin{bmatrix}1 & 0 & 0 & 0\\0 & 0 & 1 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 1\end{bmatrix}$		SWAP(qubit1 : Qubit, qubit2 : Qubit)	$SWAP\|0\rangle\|0\rangle = \|0\rangle\|0\rangle$ $SWAP\|0\rangle\|1\rangle = \|1\rangle\|0\rangle$ $SWAP\|1\rangle\|0\rangle = \|0\rangle\|1\rangle$ $SWAP\|1\rangle\|1\rangle = \|1\rangle\|1\rangle$
CNOT	$\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 1\\0 & 0 & 1 & 0\end{bmatrix}$		CNOT(control : Qubit, target : Qubit) or `(Controlled X)([control], (target));`	$CNOT\|0\rangle\|0\rangle = \|0\rangle\|0\rangle$ $CNOT\|0\rangle\|1\rangle = \|0\rangle\|1\rangle$ $CNOT\|1\rangle\|0\rangle = \|1\rangle\|1\rangle$ $CNOT\|1\rangle\|1\rangle = \|1\rangle\|0\rangle$
CCNOT, Toffoli	$\begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\\0 & 1 & 0 & 0 & 0 & 0\\0 & 0 & 1 & 0 & 0 & 0\\0 & 0 & 0 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 1\\0 & 0 & 0 & 0 & 1 & 0\end{bmatrix}$		CCNOT(control1 : Qubit, control2 : Qubit, target : Qubit) or `(Controlled X)([control1; control2], target);`	-
CSWAP, Fredkin	$\begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\\0 & 1 & 0 & 0 & 0 & 0\\0 & 0 & 1 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 1 & 0\\0 & 0 & 0 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}$		`(Controlled SWAP) ([control], (target));`	-

https://blogs.msdn.microsoft.com/uk_faculty_connection/2018/02/28/quantum-computing-quick-reference-guide/

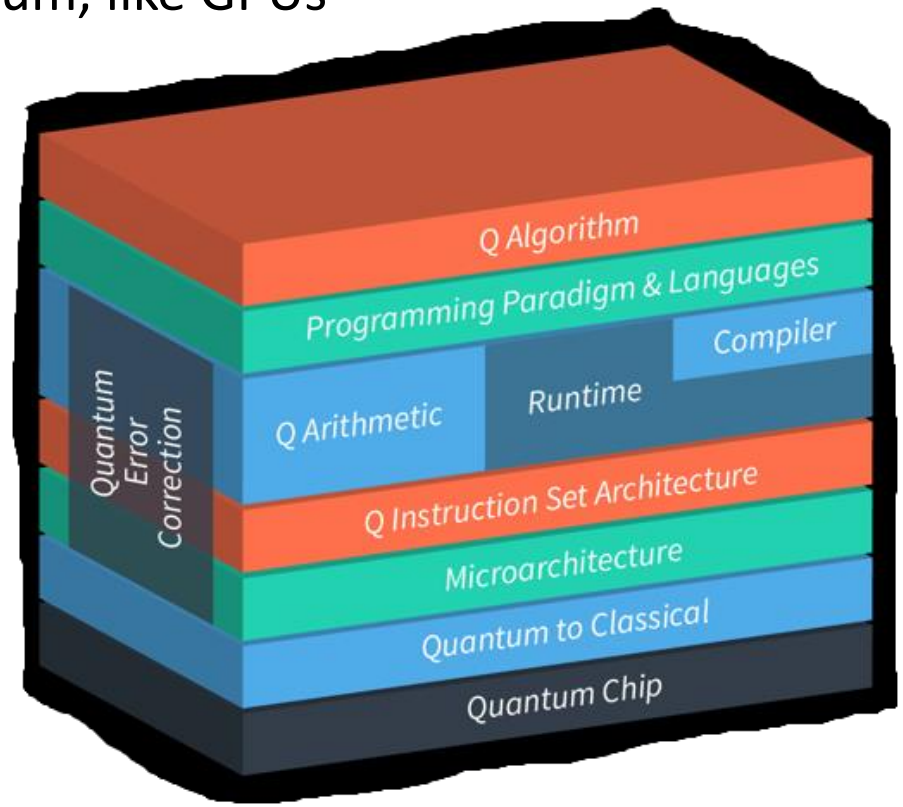
New languages & way to solve problems

- Using traditional language (Python, C/C++) or specific (Q#, Cirq, …) with additional libraries
- Using graphical editor
  - Graphical interface to program Quantum processor online
- Hybrid mode : offload specific calculation from classical to quantum, like GPUs

4 steps for a typical QC program (circuit)

1. Initialize the Quantum Computer (mostly with 0|>)
2. Encode the initial state
3. Execute the transformation / interference
4. Measure the result

Conditional branching and intermediate result readings are coming, which will make Quantum programing more like classical !



Quantum Computer as a Service (QCaaS)

Quantum Cloud	Alpine Quantum *Ion Trap*	D-Wave *Quant Annealer*	Honeywell *Ion Trap*	IBM *Superconductor*	IonQ *Ion Trap*	Quantum Ckts *Superconductor*	Rigetti *Superconductor*
AWS Quantum		⊗			⊗		⊗
Azure Quantum			⊗		⊗	⊗	
IBM Q	⊗			⊗			

➡ In addition, D-Wave, IBM, Rigetti, IonQ and QuTech offer stand-alone cloud access to their own machines.

Today, IBM Q is available for testing for **FREE** on real Quantum Computers. There are 9 different back ends (8 physical QC, and 1 simulator).

Your backends (9)

online
ibmq_16_melbourne (15 qubits)
Queue: 91 jobs

online
ibmq_burlington (5 qubits)
Queue: 4 jobs

online
ibmq_ourense (5 qubits)
Queue: 10 jobs

online
ibmq_5_yorktown - ibmqx2 (5 qubits)
Queue: 9 jobs

online
ibmq_qasm_simulator (Up to 32 qubits)
Queue: 2 jobs

online
ibmq_london (5 qubits)
Queue: 17 jobs

online
ibmq_essex (5 qubits)
Queue: 4 jobs

online
ibmq_vigo (5 qubits)
Queue: 20 jobs

online
ibmq_armonk (1 qubit)
Queue: 0 jobs

IBM Quantum Experience

File Edit Inspect View Share Help

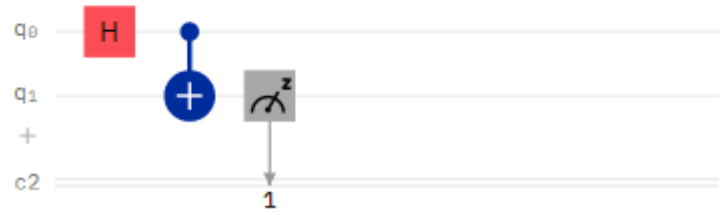
Run Settings | Run on ibmq_ourense

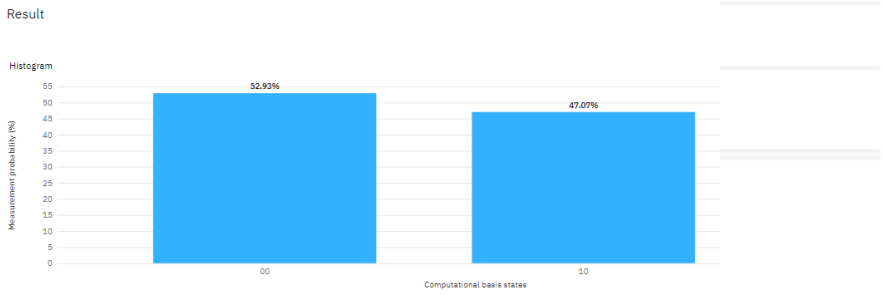
Circuits / Hello World

Code | Docs | Jobs

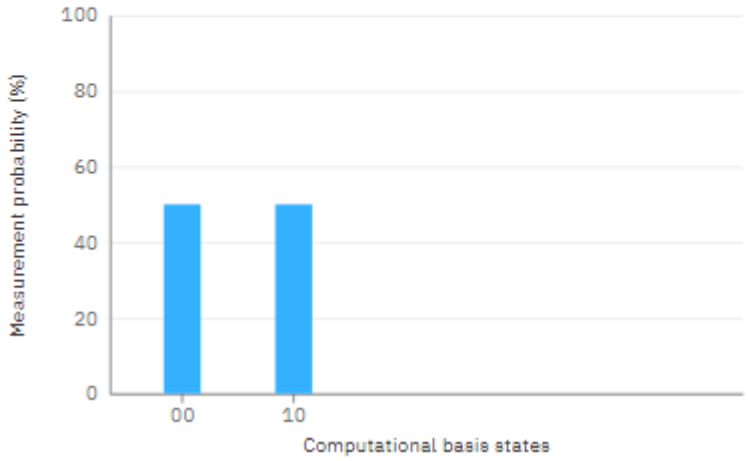
H ⊕ ⊕ ⊕ ⊕ ⊕ T S Z S† T† U1 ⋮ |0⟩ if RX RY RZ U3 Y U2 CH

CY CZ CRX CRY CRZ CU1 CU3 RXX RZZ + Add

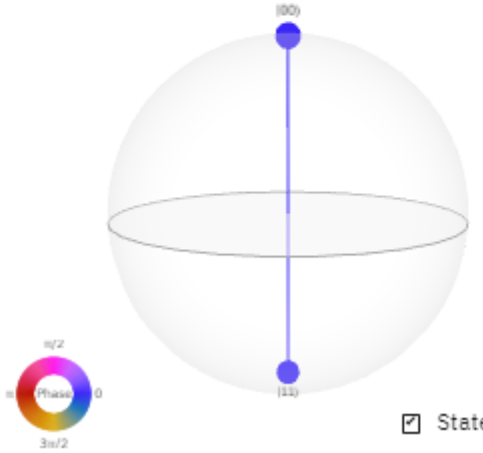




Measurement Probabilities



Q-sphere



☑ State ☐ Phase angle

Code editor

Open in Quantum Lab

QASM

```
OPENQASM 2.0;
include "qelib1.inc";

qreg q[2];
creg c[2];

h q[0];
cx q[0],q[1];
measure q[1] -> c[1];
```

Qiskit

Read only

```
from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
from numpy import pi

qreg_q = QuantumRegister(2, 'q')
creg_c = ClassicalRegister(2, 'c')
circuit = QuantumCircuit(qreg_q, creg_c)

circuit.h(qreg_q[0])
circuit.cx(qreg_q[0], qreg_q[1])
circuit.measure(qreg_q[1], creg_c[1])
```

Issues

Hardware

Decoherence / instability / noise

Almost zero-temperature for some architectures

Protection from electrical noise, radiation, etc.

Very, very difficult engineering problems

Software

Error correction

Millions of qubits needed for some applications

Using Quantum in hybrid mode

Timing

Real commercial applications probably far in time

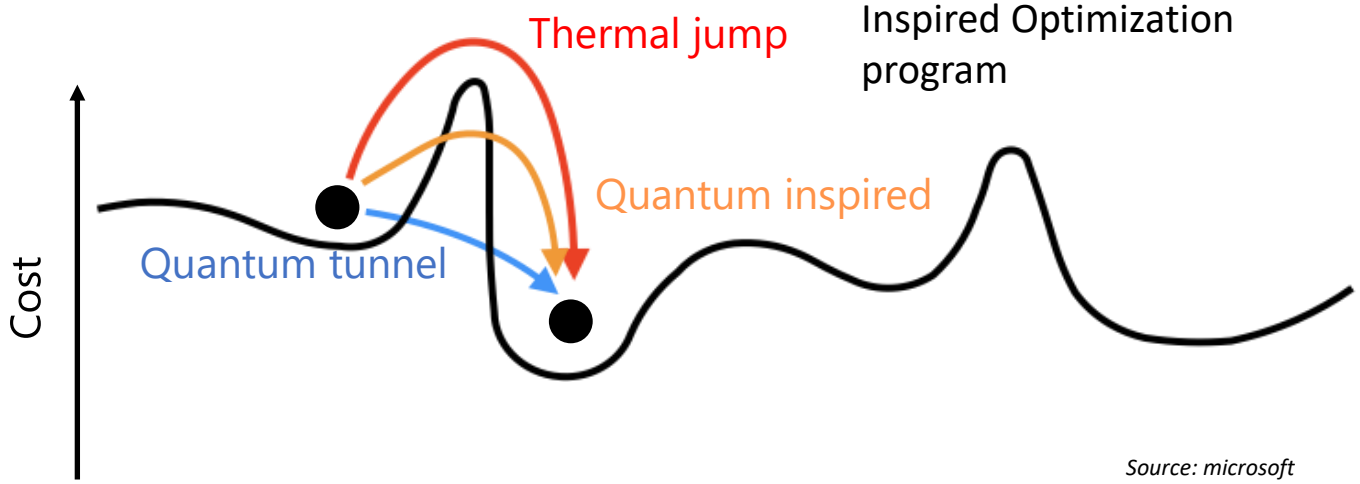
Quantum Inspired Solutions

Optimization is key in industry and science!

Quantum has mainly focused on binary problems. Why?

- Huge industrial value
- Computationally very hard
- Simple mathematical formulation
- Well suited for quantum optimizers

Example with a Quantum Inspired Optimization program



Source: microsoft

2 kind of Quantum Inspired Solutions

- Emulate quantum algorithms on classical hardware (FPGA, GPU, etc.)
- Run Quantum algorithms on physical CPU acting like quantum processor unit. (Fujitsu Digital Annealer)

Potential Applications

Short/Mid run

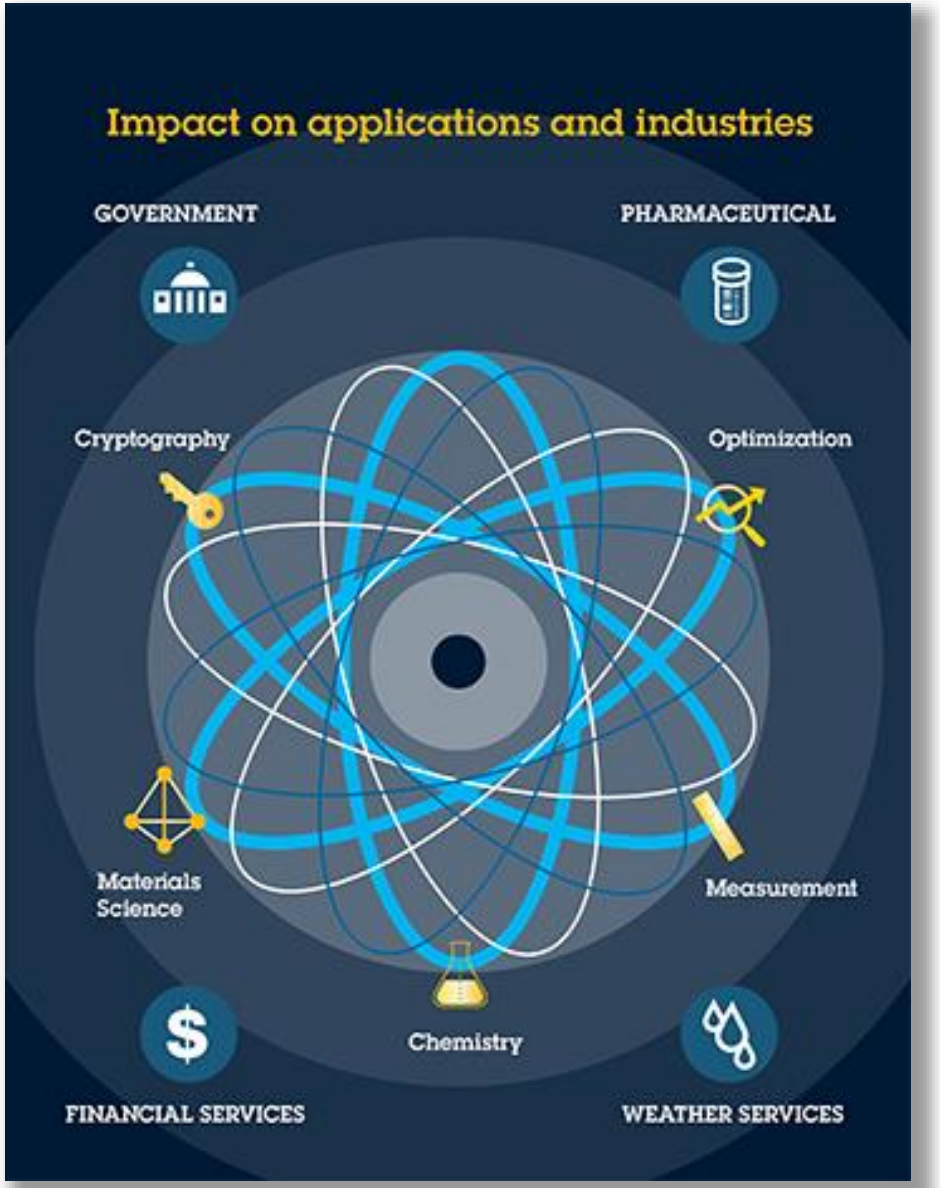
- Computing applications : in pharmaceutical research, material / chemistry science, energy, agrifood

Long run – Theoretically demonstrated

- Cryptography (Shor)
- Search problems (Grover)

Long run – Speculative

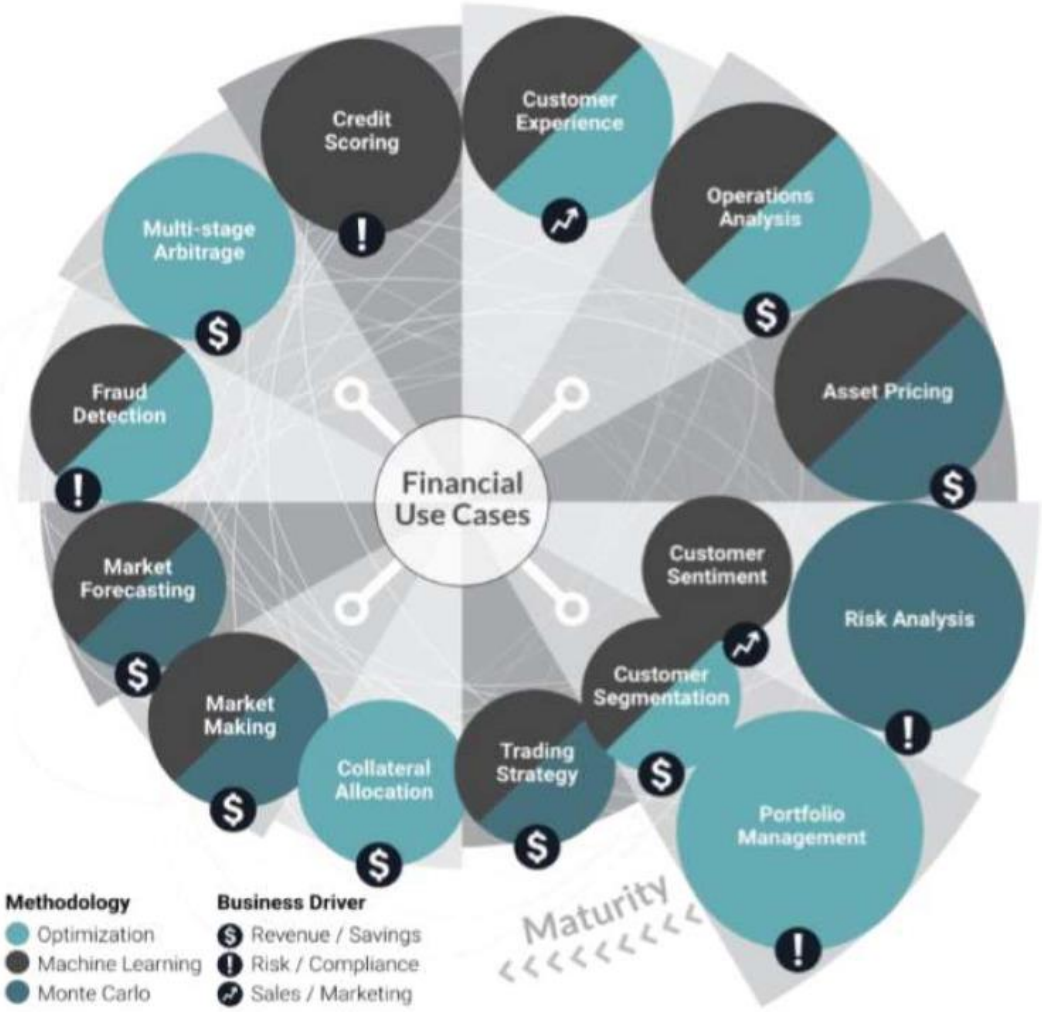
- Machine learning
  - Faster training of neural networks
  - Optimization problems
- Solving linear systems with exponential speedup



Finance use cases

	Algorithm	Application
OPTIMIZATION	• Quantum Annealing • Quantum Semi Definite Programming (QSDP) • Quantum Approximate Optimization Algorithm (QAOA)	• Portfolio optimization • Arbitration • Logistics and resource allocation • Collateral management • Market instability detection / crash prediction
SAMPLING MONTE CARLO	• Quantum Gibbs Sampling • Quantum Hamiltonian simulation • Quantum Phase estimation (QPE)	• Monte Carlo Methods • Derivative pricing • VaR/CVaR (Value at Risk) • Risk analysis
MACHINE LEARNING	• Quantum linear Systems Algorithms (QLSA, HHL) • Quantum Fourier Transform • Quantum Feature Detection (QFD) • Quantum Reinforcement Learning (QRL)	• Data modeling • Market making • Fraud / anomaly detection • Customer behavior / sentiment analysis • Prediction • Credit risk estimation

Finance use cases



Figures from: Sarah LAMOUDI, Experte IA & Quantique, #LamoudiS

France Quantum program

France has presented its Quantum plan for € 1.8 billion with 6 objectives:

- Become one of the **world leaders** in Large Scale Quantum computers (**LSQ**)
- Becoming the **European leader** in Noisy Intermediate-Scale Quantum computers (**NISQ**).
- Become one of the **world leaders in business software** based on quantum algorithms.
- Have a wide **industrial autonomy on enabling technologies** (cryogenics, lasers, ...).
- Have a wide **industrial autonomy on base sensors** of impurities in the diamond
- Maintain a **strategic independence on cryptography** technologies.

Christophe Jurczak, managing partner @Quantonation

First French Quantum Processor Unit with 100 quBits.

Technology is **based on lasers (individual atoms manipulation by laser pulse)**, this computer will run at ambient temperature, in a vacuum chamber.
It will be available for researcher and industry at the end of this year for simulation perpose (chemical, materials, …).

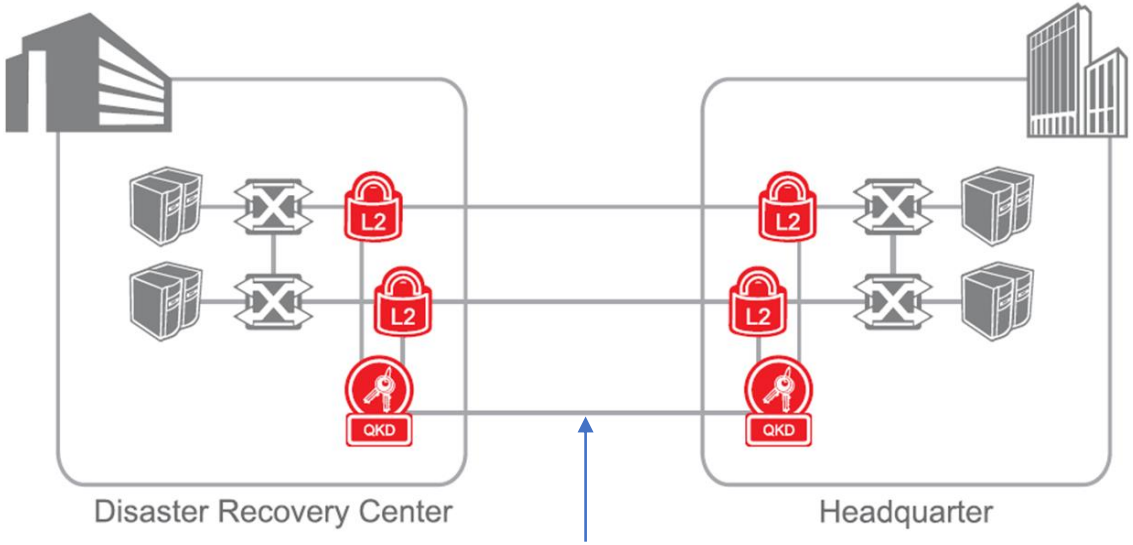
2- Quantum Cryptography

Quantum cryptography

We can exploit the eavesdropper detection that is intrinsic to quantum systems (**no-copy theory**) in order to derive new “unconditionally secure” information security protocols. The **security depends only on the laws of physics**, and not on computational assumptions.

Quantum key Distribution (**QKD**)

(BB84 protocol : C. Bennet & G. Brassard in 1984)



Dedicated fiber connection to exchange cryptographic keys (BB84)

Using quantum physics to exchange encryption keys, and then using these keys to encrypt the data over a traditional communication link

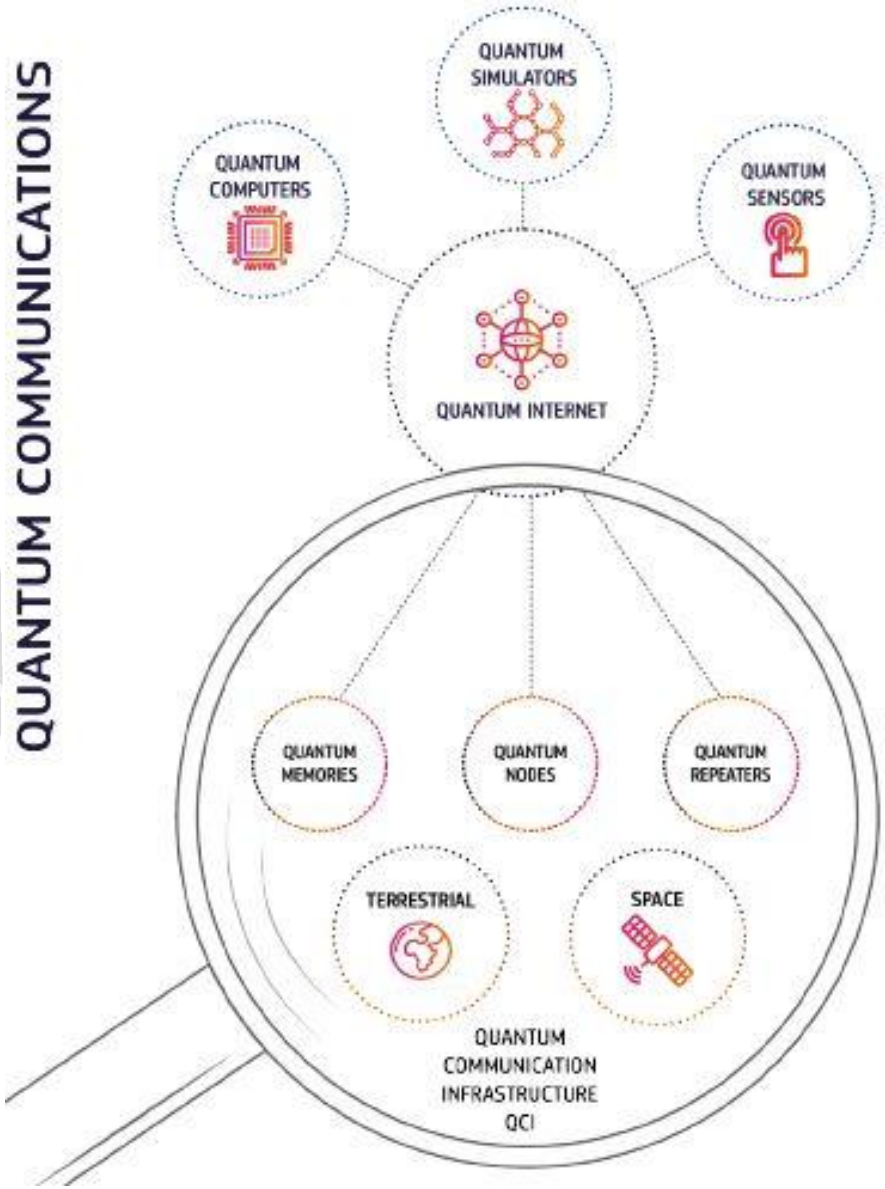
Quantum secured network

QUANTUM FLAGSHIP

European Commission

20 EU countries agree to work together to explore how to build a Quantum Communication Infrastructure (QCI)

QUANTUM COMMUNICATIONS



China has developed the world's first mobile quantum satellite station (Q1 2020).

Unlike the ground station used when Mozi (Chinese Quantum Satelite) launched (in 2017), which weighed more than **10 tonnes**, the **mobile station weighs about 80 kilograms** and is small enough to be installed on top of a car. **A key was relayed via Mozi between the mobile ground station** in Jinan and a fixed station in Shanghai. **ICBC and the People's Bank of China** are already **using satellite-based quantum key distribution between distant cities**, such as Beijing in north-east China and Urumqi in the far north-west.

Quantum Random Number Generator

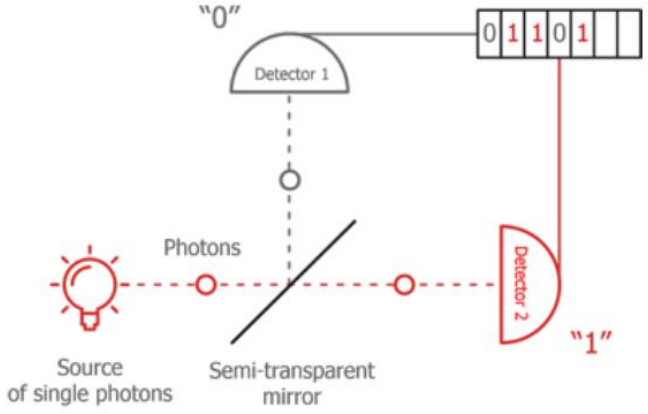
Random number are used everywhere to **secure communication / datas**

- SSL protocol
- Wifi WPA2 protocol
- Sending encrypted emails
- Skype video chats
- …

A lot of security breaches are the result of using **pseudo random numbers**.

"Random number generation is too important to be left to chance."
(R.R. Coveyou, 1970 [1])

Certifiable randomness has many potential applications, for example to secure communication protocols, unbiased statistical sampling, and Monte Carlo simulations.

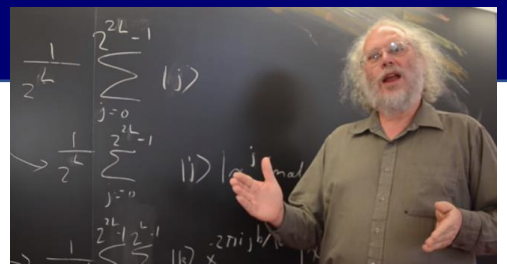


Schematics of a QRNG based on single-photon splitting
Randomness is based on physical laws of light duality

"What source of random should I use" has a simple, safe and commercially viable answer**:**
Use Quantum!
(source: Cloud Security Alliance.org)

3- Post-Quantum Cryptography

Shor's factoring algorithm



The result of Shor's factoring algorithm applied on different RSA key length:

With a classical computer

# bits	1024	2048	4096
factoring in 2006	10^5 years	$5x10^{15}$ years	$3x10^{29}$ years
factoring in 2024	38 years	10^{12} years	$7x10^{25}$ years
factoring in 2042	3 days	$3x10^8$ years	$2x10^{22}$ years

With potential quantum computer

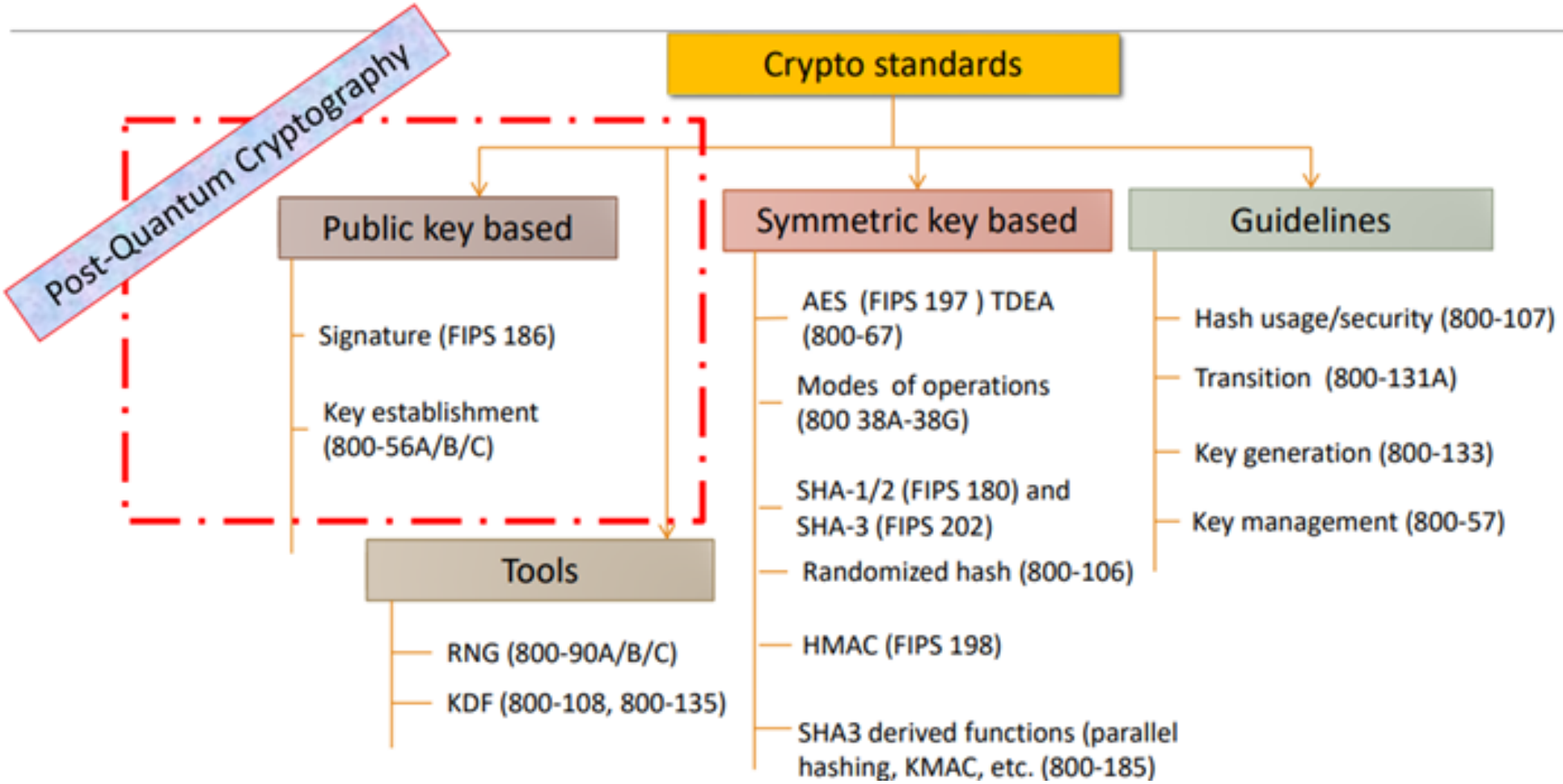
(e.g., clock speed 100 MHz)

# bits	1024	2048	4096
# qubits	5124	10244	20484
# gates	$3x10^9$	$2X10^{11}$	$X10^{12}$
factoring time	4.5 min	36 min	4.8 hours

"Based on this, **the RSA / ECC like keys will be no more secure** when the Quantum computer will be reality. ", ***R. J. Hughes, LA-UR-97-4986***

Shor's factoring algorithm Impact

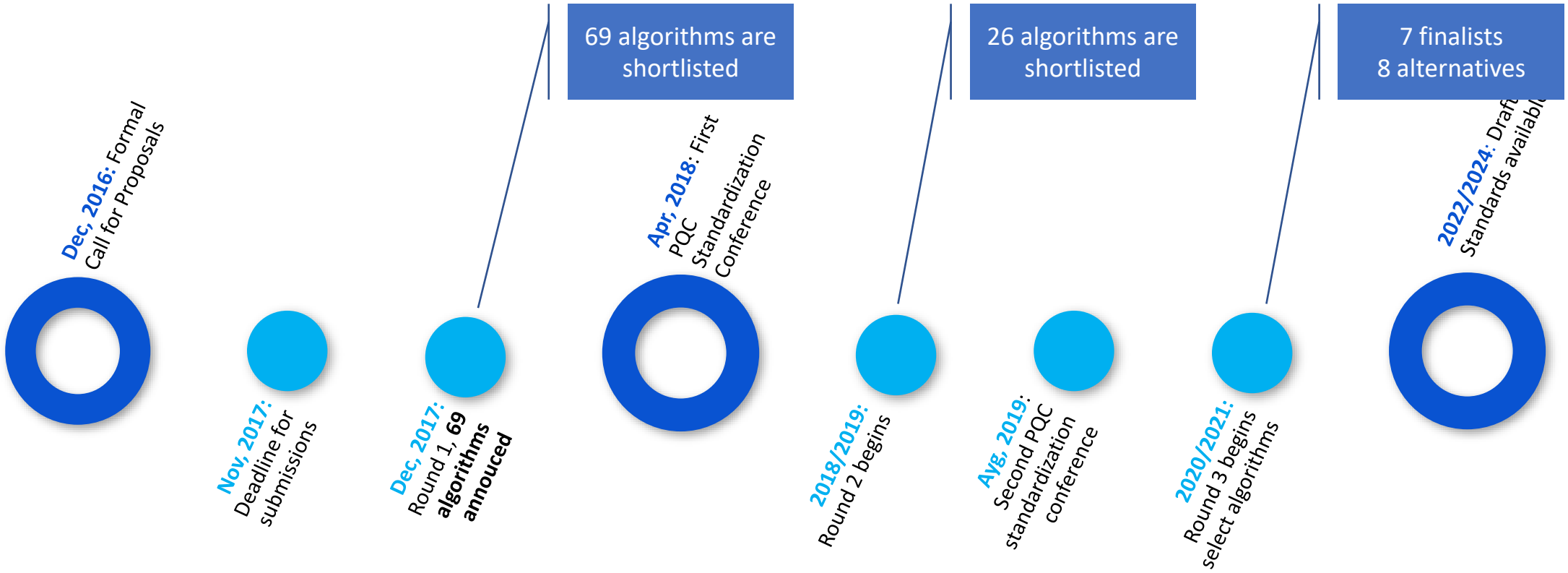
Quantum Impact to NIST Standards



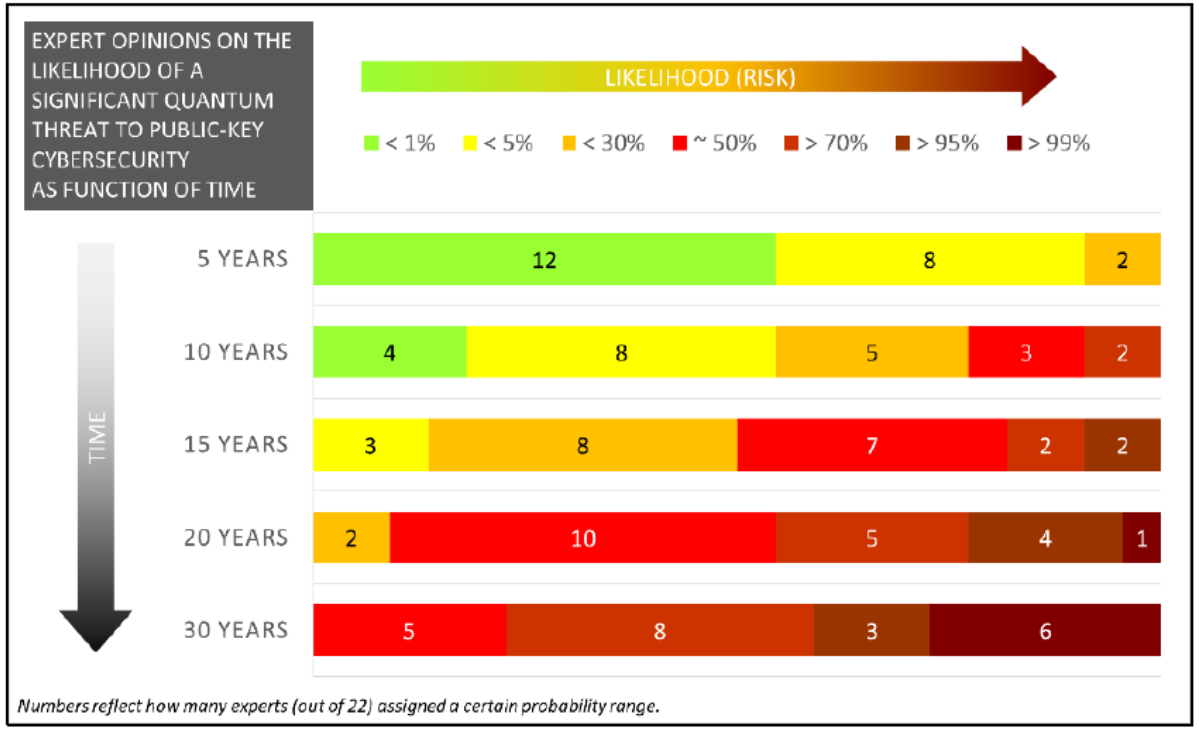
Post-quantum cryptography

Post-quantum cryptography is the domain in public-key cryptography, to find algorithm which will be resistant to the power of quantum computers.

NIST (National Institute of Standards and Technology) has initiated a process to solicit, evaluate, and standardize one or more quantum-resistant public-key cryptographic algorithms.



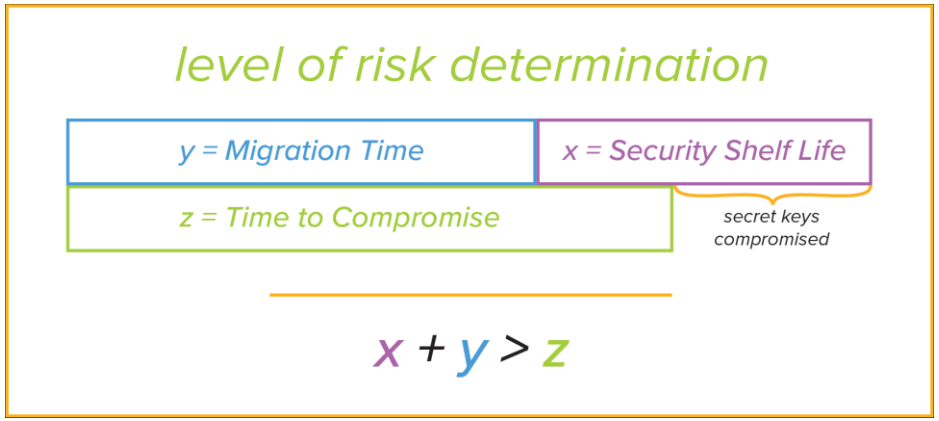
Post-quantum cryptography



Sources: Global Risk Institute (https://globalriskinstitute.org/)

There is an inflection point for the experts **in 15 to 20 years from now**, where the **quantum threat** has a "roughly 50%" or greater chance of materializing.

And 30 years from now, there will be, for the experts, no doubt.



Given the breadth and complexity of the transition ahead, it is essential for companies to **begin transitioning now.**

“If x+y > z, you have a serious problem today !” (Mosca’s inegality)

Take away

- There are 2 kinds of Quantum Computer
  - **Digital / Universal** : IBM, Google, Microsoft, Rigetti, IonQ, Honeywell, …
  - **Analogic / specialized** : D-Wave, simulators (Fujitsu Digital Annealer (simulator), Toshiba

- The first usable **noisy quantum computer (NISQ)** could be available in **3 to 5 years**, where the **error free quantum computer** could be available in **10 to 20 years** from now.
  - **It is a very, very, very complex subject and a lot of engineering problems to solve**

- **It is possible to take advantage from Quantum computing already today with Quantum Inspired Solutions**

- Solution taking advantage of quantum physics effects **are available**
  - Quantum Key Distribution (**QKD**)
  - Quantum Random Number Generator (**QRNG**)

- Not all cryptography protocol are vulnerable, and cryptographic algorithm resistant to Quantum Supremacy are under standardization **(post-quantum cryptography),** it's time to include quantum threats in the **cybersecurity risk assessment**

And for the end, **Just for the skeptical**

- I think there can be a world market for **maybe five computers**. (Thomas Watson, CEO of IBM, 1943)
- There is **no reason** for an individual to have a computer at home . (Ken Olsen , president, director and founder of Digital Equipment Corp., 1977)
- I think that this thing (WWW) that Tim (Berners-Lee) has shown me **has no future** (F. Carminati, 1989)

APPENDICES

Quantum Inspire - By QuTech

Quick Guide FAQ Close editor

Editor Results

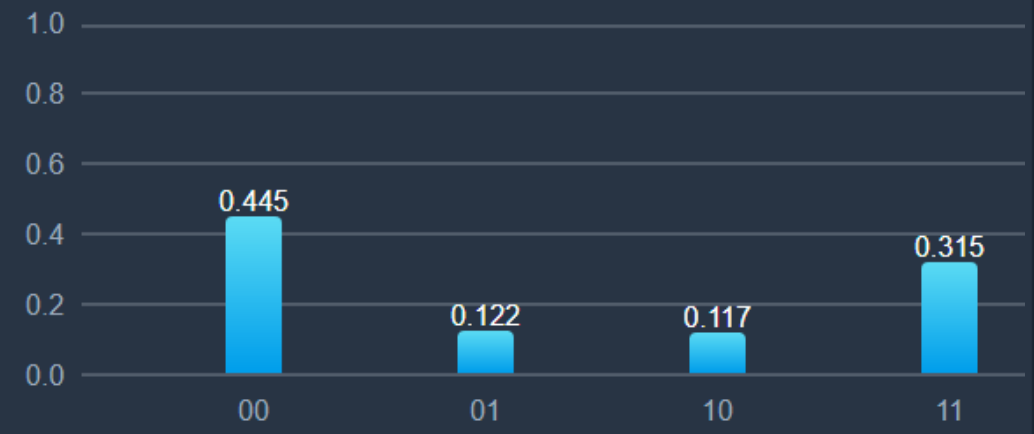
Hello World

Saved Select backend Run

```
version 1.0

qubits 2

#hello world
# initialize the 2 qbits
{ prep_z q[0] | prep_z q[1] }
# Superposition
H q[0]
# Entanglement
CNOT q[0],q[1]
```



Operations | Backend Information

Spin-2

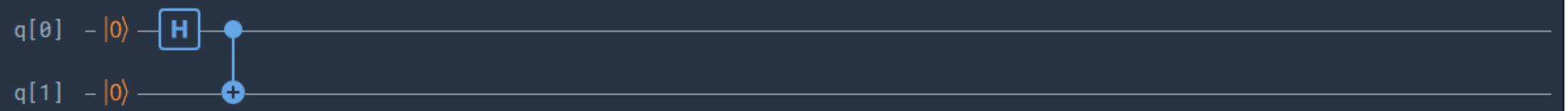
Backend status: Idle

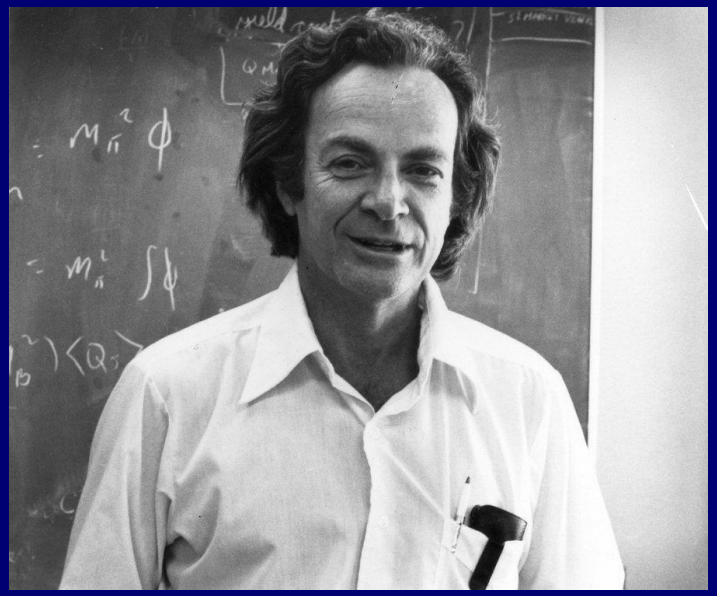
Fridge temperature: 42mK

Calibration date: 10/14/2020 - 4:16:17 PM

Learn more

	ω (GHz)	T2* (μs)	Vis (%)
q0	15.61	3.1	87.8
q1	15.83	7.8	58.2





Richard Feynman

1918-1988
Nobel Prize in Physics (quantum electrodynamics) in 1965

“I think I can safely say that nobody understands quantum mechanics”